```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

data = pd.read_csv(r"C:\Users\v.rajkumar\Downloads\Day-20\archive (2)\
ai_job_market_insights.csv")

data.head()
```

	Job_Title	Industry	Company_Size	Location	AI_Adoption_Level	Automation_Risk	Required_Skills	Salary_USD	Remote_Friendly	Job_Growth_Projection
0	Cybersecurity Analyst	Entertainment	Small	Dubai	Medium	High	UX/UI Design	111392.165243	Yes	Growth
1	Marketing Specialist	Technology	Large	Singapore	Medium	High	Marketing	93792.562466	No	Decline
2	AI Researcher	Technology	Large	Singapore	Medium	High	UX/UI Design	107170.263069	Yes	Growth
3	Sales Manager	Retail	Small	Berlin	Low	High	Project Management	93027.953758	No	Growth
4	Cybersecurity Analyst	Entertainment	Small	Tokyo	Low	Low	JavaScript	87752.922171	Yes	Decline

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Job_Title             500 non-null    object
 1   Industry              500 non-null    object
 2   Company_Size          500 non-null    object
 3   Location              500 non-null    object
 4   AI_Adoption_Level     500 non-null    object
 5   Automation_Risk       500 non-null    object
```

```
 6   Required_Skills        500 non-null    object
 7   Salary_USD             500 non-null    float64
 8   Remote_Friendly        500 non-null    object
 9   Job_Growth_Projection  500 non-null    object
dtypes: float64(1), object(9)
memory usage: 39.2+ KB
```

```
data.describe()
```

```
          Salary_USD
count     500.000000
mean    91222.390974
std     20504.291453
min     31969.526346
25%     78511.514863
50%     91998.195286
75%    103971.282092
max    155209.821614
```

```
data.isnull().sum()
```

```
Job_Title                0
Industry                 0
Company_Size             0
Location                 0
AI_Adoption_Level        0
Automation_Risk          0
Required_Skills          0
Salary_USD               0
Remote_Friendly          0
Job_Growth_Projection    0
dtype: int64
```

```
data.duplicated().sum()
```

```
0
```

```
data.columns
```

```
Index(['Job_Title', 'Industry', 'Company_Size', 'Location',
       'AI_Adoption_Level', 'Automation_Risk', 'Required_Skills',
'Salary_USD',
       'Remote_Friendly', 'Job_Growth_Projection'],
      dtype='object')
```

```
data["Job_Title"].unique()
```

```
array(['Cybersecurity Analyst', 'Marketing Specialist', 'AI
Researcher',
       'Sales Manager', 'UX Designer', 'HR Manager', 'Product
Manager',
```

```
       'Software Engineer', 'Data Scientist', 'Operations Manager'],
      dtype=object)
```

```
data["Industry"].unique()
```

```
array(['Entertainment', 'Technology', 'Retail', 'Education',
'Finance',
       'Transportation', 'Telecommunications', 'Manufacturing',
       'Healthcare', 'Energy'], dtype=object)
```

```
data["Company_Size"].unique()
```

```
array(['Small', 'Large', 'Medium'], dtype=object)
```

```
data["AI_Adoption_Level"].unique()
```

```
array(['Medium', 'Low', 'High'], dtype=object)
```

```
data["Location"].unique()
```

```
array(['Dubai', 'Singapore', 'Berlin', 'Tokyo', 'San Francisco',
'London',
       'Paris', 'Sydney', 'New York', 'Toronto'], dtype=object)
```

```
data["Automation_Risk"].unique()
```

```
array(['High', 'Low', 'Medium'], dtype=object)
```

```
data["Required_Skills"].unique()
```

```
array(['UX/UI Design', 'Marketing', 'Project Management',
'JavaScript',
       'Cybersecurity', 'Sales', 'Machine Learning', 'Python',
       'Data Analysis', 'Communication'], dtype=object)
```

```
data["Salary_USD"].unique()
```

```
array([111392.16524316,  93792.56246611, 107170.26306895,
93027.95375786,
        87752.92217059, 102825.00786657, 102065.72067348,
86607.3176181 ,
        75015.86084571,  96834.57829481,  91566.97337925,
78902.56574458,
        73151.99034926,  98209.51687789, 101001.90114055,
82799.3577069 ,
        78401.25928174,  73394.89621926,  97645.0987829 ,
73920.45852719,
       134822.69138041,  81794.37128058,  41495.71444255,
89396.56666742,
        88539.33040362, 112234.37135048,  95762.52473455,
73323.2963312 ,
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```

```
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86711.67119077,
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```

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```

```
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105821.39404552,
       119794.99214641,  79644.93309909,  77642.15062535,
68764.3789208 ])

data["Remote_Friendly"].unique()

array(['Yes', 'No'], dtype=object)

data["Job_Growth_Projection"].unique()

array(['Growth', 'Decline', 'Stable'], dtype=object)

data.columns
```

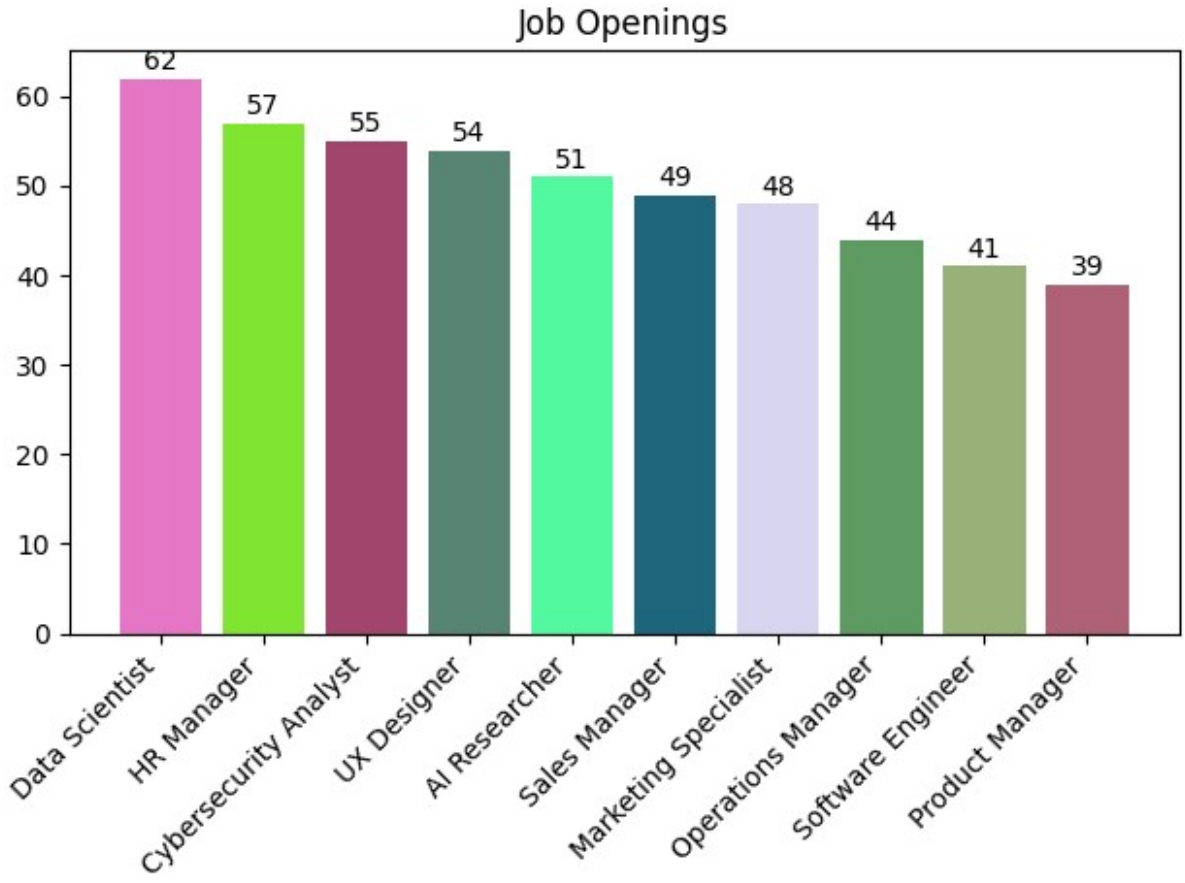
```
Index(['Job_Title', 'Industry', 'Company_Size', 'Location',
       'AI_Adoption_Level', 'Automation_Risk', 'Required_Skills',
'Salary_USD',
       'Remote_Friendly', 'Job_Growth_Projection'],
      dtype='object')
```

```
## **Visualize**

job_count = data["Job_Title"].value_counts()

plt.title("Job Openings")
plt.bar(job_count.index,job_count.values,color=np.random.rand(10,3))
plt.xticks(rotation=45,ha="right")

for i,v in enumerate(job_count.values):
    plt.text(i,v+1,str(v),ha="center")
plt.tight_layout()
```

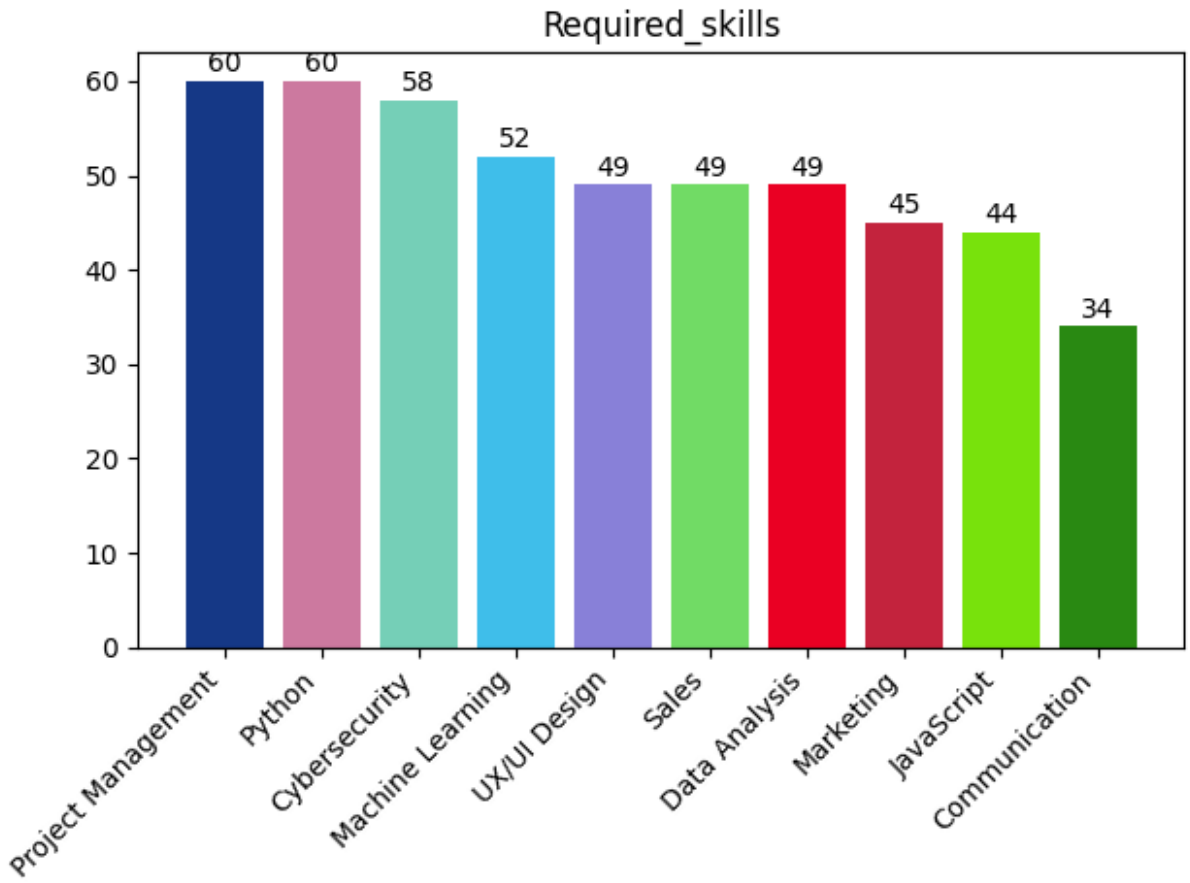



```
 rs = data["Required_Skills"].value_counts()

plt.title("Required_skills")
plt.bar(rs.index,rs.values,color=np.random.rand(10,3))
```

```
plt.xticks(rotation=45,ha="right")

for i,v in enumerate(rs.values):
    plt.text(i,v+1,str(v),ha="center")
plt.tight_layout()
```



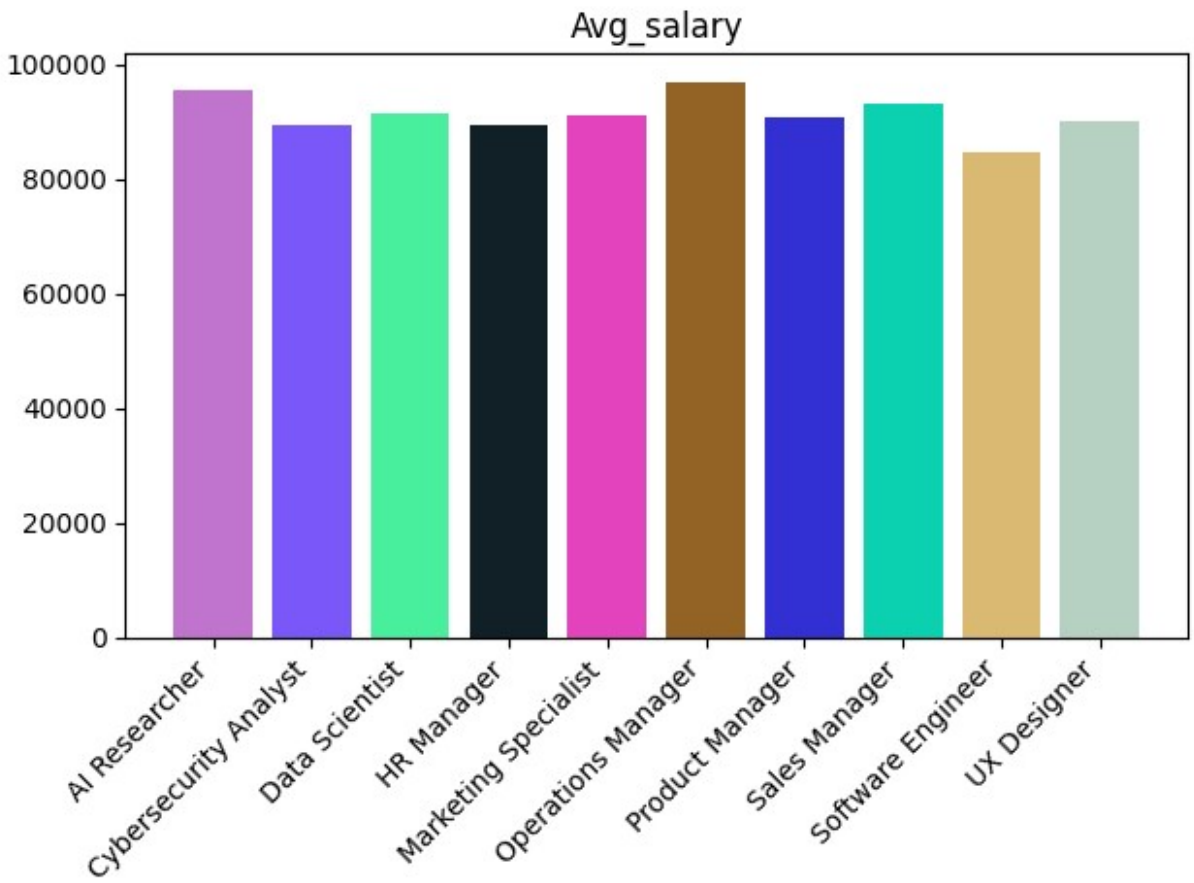

```
data.columns

Index(['Job_Title', 'Industry', 'Company_Size', 'Location',
       'AI_Adoption_Level', 'Automation_Risk', 'Required_Skills',
'Salary_USD',
       'Remote_Friendly', 'Job_Growth_Projection'],
      dtype='object')

avg_salary = data.groupby("Job_Title")["Salary_USD"].mean()

plt.title("Avg_salary")
plt.bar(avg_salary.index,avg_salary.values,color=np.random.rand(10,3))
plt.xticks(rotation=45,ha="right")

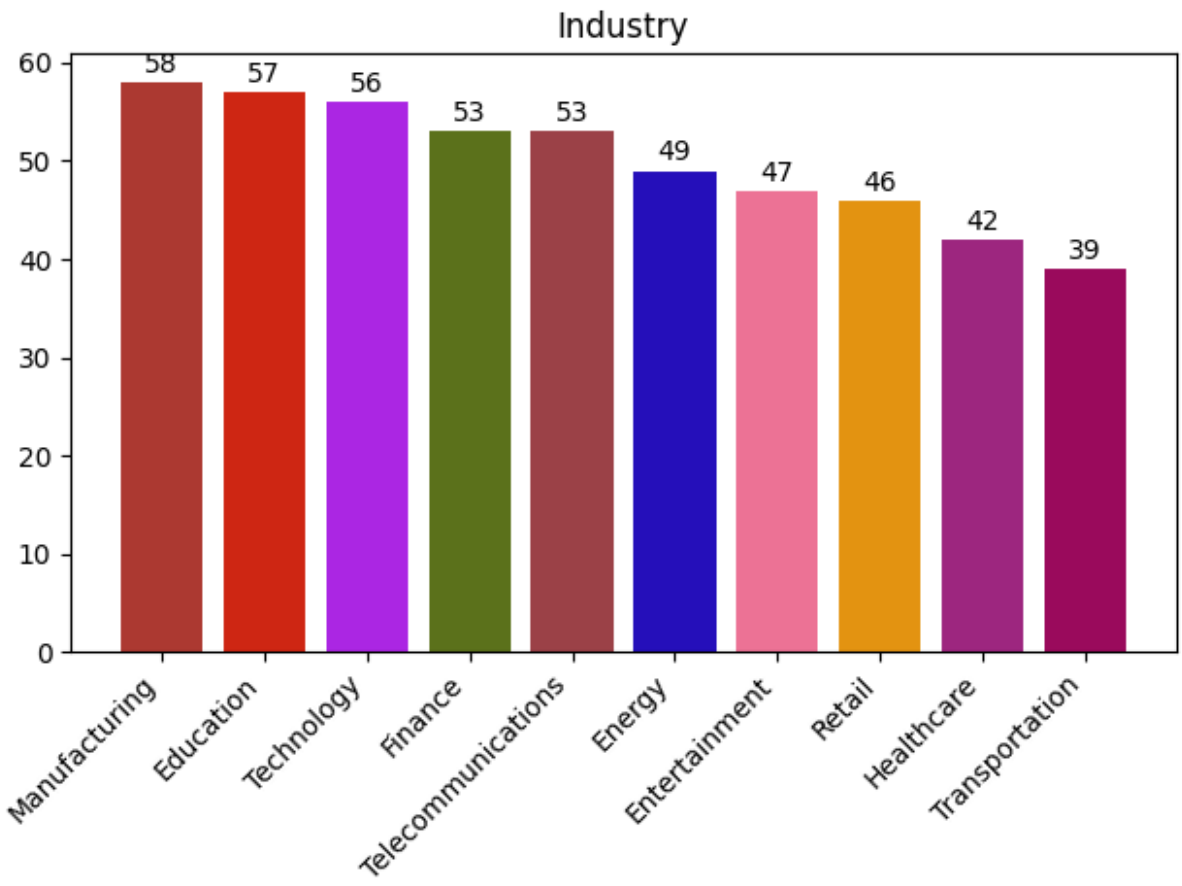
plt.tight_layout()
```




```
indus = data["Industry"].value_counts()

plt.title("Industry")
plt.bar(indus.index,indus.values,color=np.random.rand(10,3))
plt.xticks(rotation=45,ha="right")

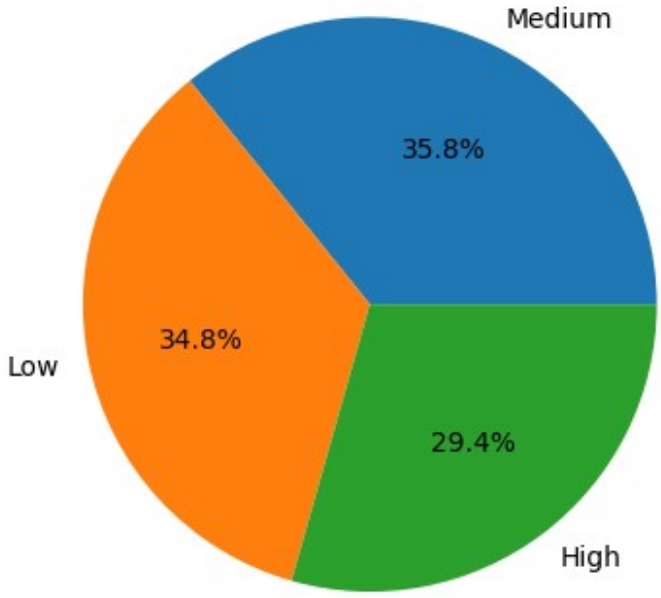
for i,v in enumerate(indus.values):
    plt.text(i,v+1,str(v),ha="center")
plt.tight_layout()
```




```
ai_adapt_level = data["AI_Adoption_Level"].value_counts()

plt.pie(ai_adapt_level.values,labels=ai_adapt_level.index,autopct="%.1
f%%")
```

```
([<matplotlib.patches.Wedge at 0x1d65f289cc8>,
  <matplotlib.patches.Wedge at 0x1d65f28cc08>,
  <matplotlib.patches.Wedge at 0x1d65f283588>],
 [Text(0.47460161828792374, 0.9923473705908048, 'Medium'),
  Text(-1.0778405503827289, -0.21968101408782775, 'Low'),
  Text(0.6632224965152421, -0.8775738829956655, 'High')],
 [Text(0.2588736099752311, 0.5412803839586208, '35.8%'),
  Text(-0.5879130274814884, -0.11982600768426967, '34.8%'),
  Text(0.3617577253719502, -0.47867666345218113, '29.4%')])
```




```
location = data["Location"].value_counts()

plt.title("Location based openings")
plt.bar(indus.index,indus.values,color=np.random.rand(10,3))
plt.xticks(rotation=45,ha="right")

for i,v in enumerate(indus.values):
    plt.text(i,v+1,str(v),ha="center")
plt.tight_layout()
```